Vidya Nirmaan

AI-Powered School Infrastructure Planning & Monitoring System

Architecture & Design Document

IndiaAI Innovation Challenge

Problem Statement 5: School Infrastructure Management

Department of School Education, Andhra Pradesh

Version	1.0
Date	March 2026
Platform	Flutter (iOS + Android) + Supabase + FastAPI
Coverage	319 Schools, 57 Districts, 707 Mandals

Table of Contents

1.	Executive Summary	3
2.	System Architecture	4
3.	Technology Stack	5
4.	Database Schema (ER Diagram)	6
5.	AI/ML Pipeline: Priority Scoring	7
6.	AI/ML Pipeline: Demand Validation	8
7.	AI/ML Pipeline: Enrolment Forecasting	9
8.	3-Stage Demand Approval Pipeline	10
9.	Role-Based Access Control	11
10.	Offline-First Architecture	12
11.	Screen Navigation & User Flow	13

1. Executive Summary

Problem Statement

Andhra Pradesh manages over 45,000+ government schools across 57 districts and 707 mandals. The current infrastructure planning process for facilities like CWSN resource rooms, accessible toilets, drinking water, electrification, and ramps relies on manual paper-based workflows prone to data inconsistencies, delayed processing, fraudulent demand inflation, and inequitable resource allocation. There is no automated system to validate infrastructure demands, forecast future needs based on enrolment trends, or prioritize schools objectively.

Our Solution: Vidya Nirmaan

Vidya Nirmaan is an AI-powered mobile application that digitizes the entire school infrastructure lifecycle: from demand raising by Head Masters, to AI-driven validation and anomaly detection, field assessment by inspectors, and final officer approval. The system uses machine learning models including Isolation Forest for fraud detection, Linear Regression and Cohort Progression for enrolment forecasting, and a weighted composite scoring algorithm for objective school prioritization.

Key Metrics

Metric	Value
Schools Covered	319 (pilot), scalable to 45,000+
Districts	57
Mandals	707
Demand Plans Managed	799
Enrolment Records	4,638
Infrastructure Types	5 (Samagra Shiksha norms)
AI Models	3 (Priority Scoring, Validation, Forecasting)
Validation Rules	7 rule-based + Isolation Forest ML
User Roles	5 (State, District, Block, Inspector, HM)
Languages	2 (English + Telugu)
Offline Support	Full offline-first with auto-sync

2. System Architecture

Vidya Nirmaan follows a 3-tier architecture: a Flutter mobile frontend, a Supabase backend (PostgreSQL with Row Level Security), and an optional FastAPI ML backend deployed on Railway cloud. A Hive-based local storage layer provides offline-first capability with automatic synchronization.

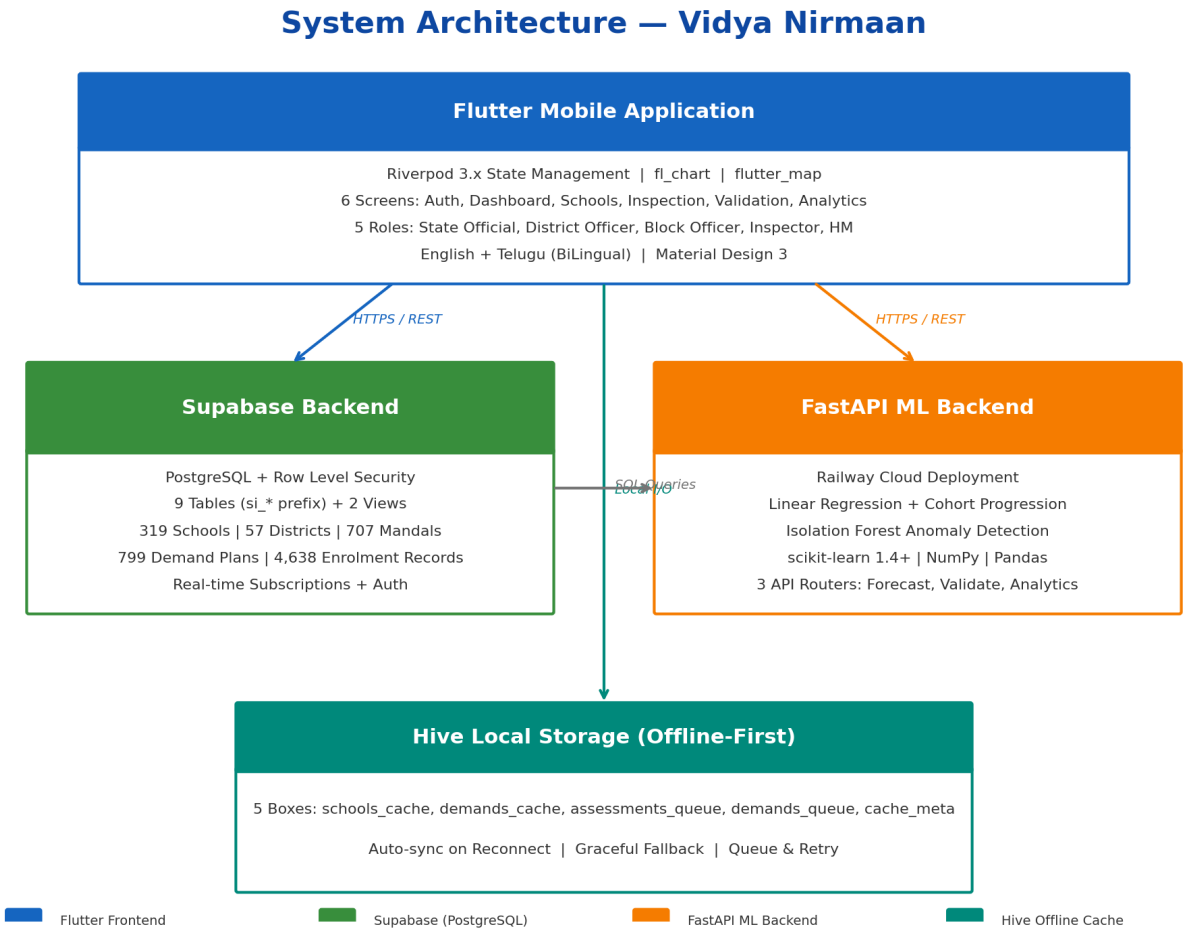

Figure 1: System Architecture — 3-Tier with Offline-First Design

3. Technology Stack

Layer	Technology	Purpose
Frontend	Flutter 3.10+	Cross-platform mobile (iOS + Android)
State Management	Riverpod 3.x	Reactive state with Notifier/AsyncValue pattern
Charts	fl_chart	Enrolment charts, priority pie, analytics
Maps	flutter_map	School location visualization, priority-coded markers
Backend DB	Supabase	PostgreSQL + Auth + Row Level Security
ML Backend	FastAPI 0.109+	Python ML microservice
ML Cloud	Railway	FastAPI deployment and hosting
ML Libraries	scikit-learn 1.4+	LinearRegression, IsolationForest
Data Science	NumPy, Pandas	Data processing and feature engineering
Local Storage	Hive	Offline-first caching (5 boxes)
HTTP Client	Dio	REST API communication with timeout/retry
Export	pdf, excel	PDF and Excel report generation
Localization	Custom i18n	English + Telugu bilingual support
UI Framework	Material Design 3	Modern, accessible UI components

Infrastructure Types & Unit Costs (Samagra Shiksha Norms)

Infrastructure Type	Unit Cost (Lakhs)	Description
CWSN Resource Room	29.30	Special needs classroom facility
CWSN Toilet	4.65	Accessible toilet block for differently-abled
Drinking Water	3.40	Water purification and supply system
Electrification	1.75	School electrical infrastructure
Ramps	1.25	Accessibility ramps and handrails

4. Database Schema (Entity-Relationship Diagram)

All tables use the **si_** prefix (school infrastructure). The schema follows a hierarchical geographic structure: Districts → Mandals → Schools, with satellite tables for enrolment history, demand plans, infrastructure assessments, priority scores, and ML forecasts. Two database views (*si_schools_view* and *si_demand_plans_view*) provide pre-joined data for efficient querying.

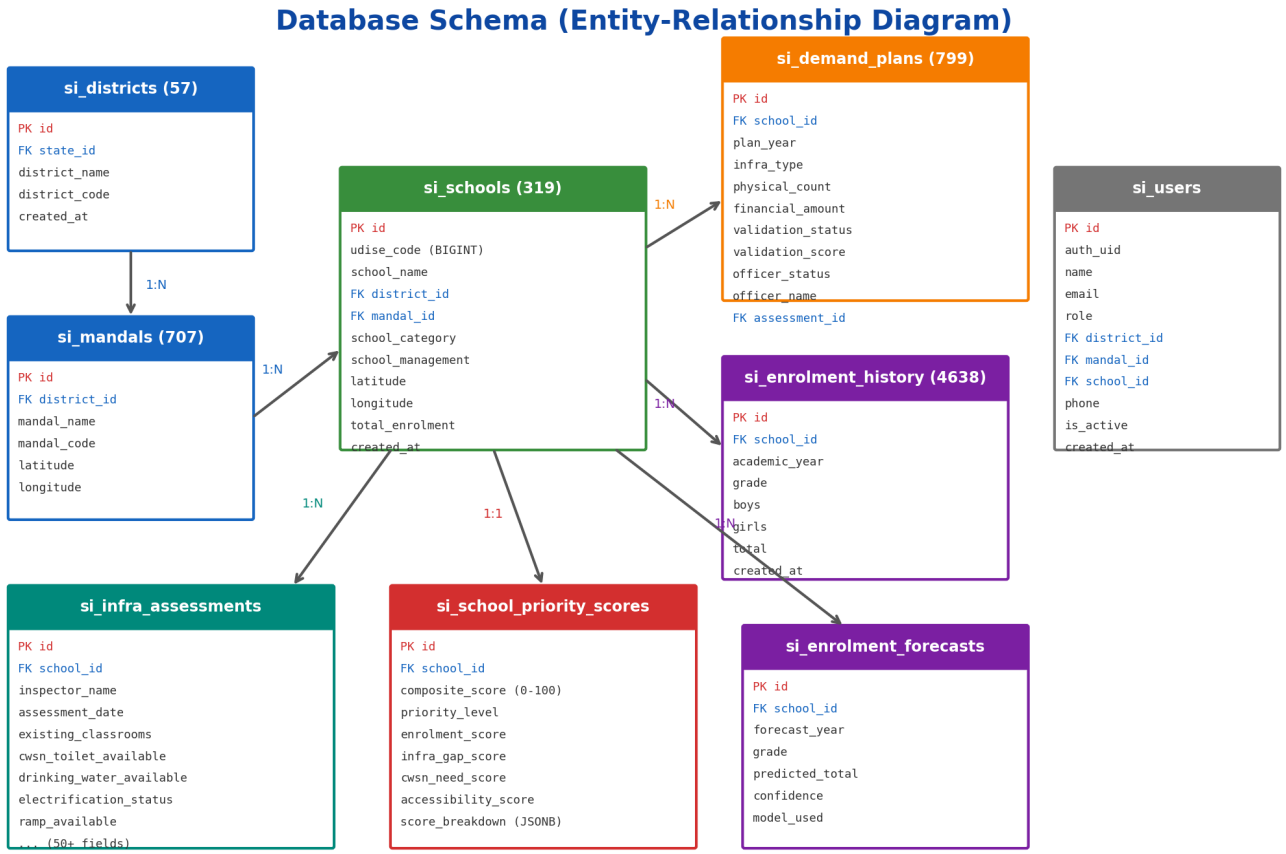

Figure 2: Entity-Relationship Diagram — 9 Tables with Geographic Hierarchy

5. AI/ML Pipeline: Priority Scoring

The Priority Scoring algorithm computes a composite score (0-100) for each school using four weighted factors. This score objectively ranks schools for infrastructure investment, replacing subjective manual assessment. Scores are computed client-side for instant results and stored in the **si_school_priority_scores** table.

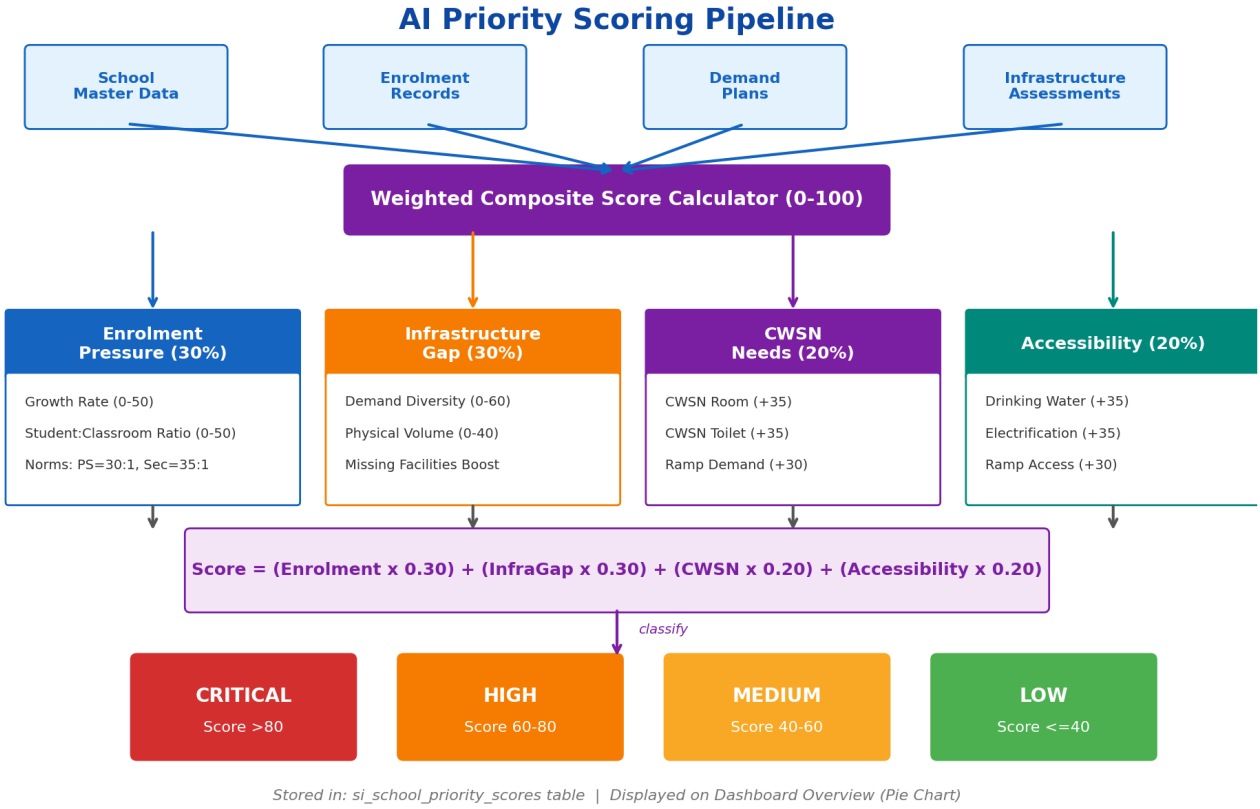

Figure 3: Priority Scoring Pipeline — 4-Factor Weighted Composite (0-100)

6. AI/ML Pipeline: Demand Validation

Infrastructure demands pass through a dual-layer validation engine. The client-side rule-based engine applies 7 validation checks with cumulative score deductions. The backend ML layer uses an **Isolation Forest** anomaly detector trained on the demand dataset to identify statistical outliers that rules alone may miss. Demands scoring ≥80 are auto-approved, 50-79 are flagged for review, and <50 are rejected.

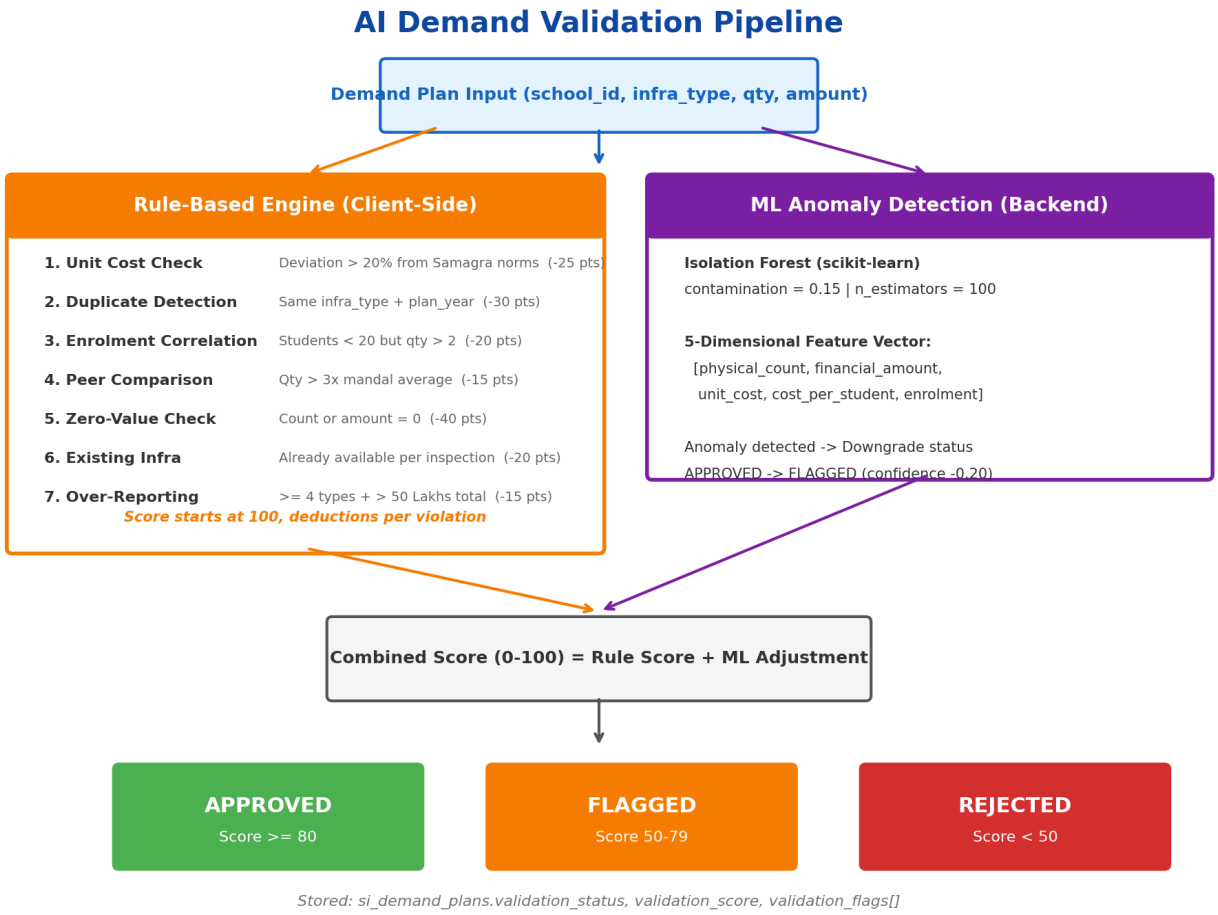

Figure 4: Demand Validation Pipeline — 7 Rules + Isolation Forest ML

7. AI/ML Pipeline: Enrolment Forecasting

Enrolment forecasting uses a dual-model approach: **Linear Regression** for overall trend extrapolation and **Cohort Progression** for grade-wise student flow tracking. A multi-factor confidence score accounts for data quality, prediction horizon decay, and historical volatility. The system follows a backend-first strategy, falling back to client-side linear extrapolation when the ML backend is unavailable.

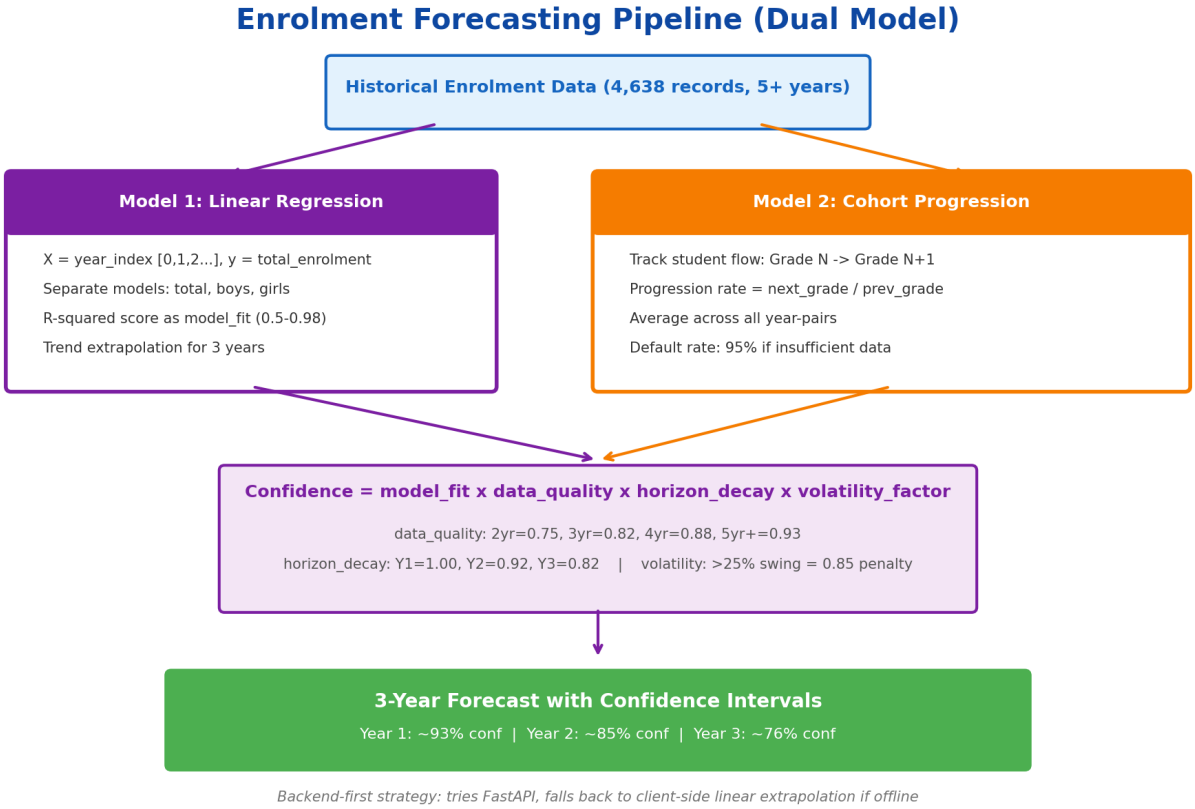

Figure 5: Enrolment Forecasting — Dual Model with Confidence Decay

8. 3-Stage Demand Approval Pipeline

Every infrastructure demand passes through three sequential stages: **AI Validation** (automated scoring and anomaly detection), **Field Assessment** (physical inspection by field officers with 50+ data points and GPS verification), and **Officer Decision** (manual approval by District/State officials). A **hard gate** enforces that field assessment must be completed before officer approval, ensuring no demand is approved without ground-truth verification.

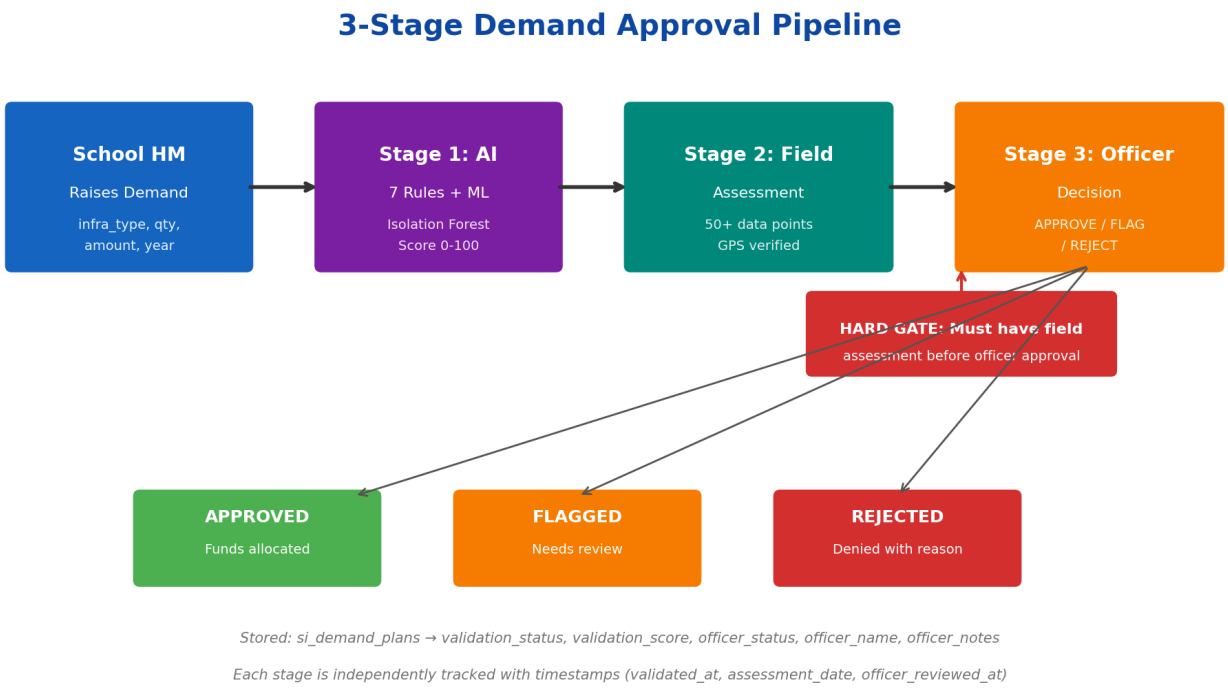

Figure 6: 3-Stage Approval Pipeline — AI, Assessment, Officer with Hard Gate

9. Role-Based Access Control

The application implements strict role-based access control with 5 distinct user roles. Data scoping is enforced at the provider level using **effectiveDistrictProvider**, **effectiveMandalProvider**, and **effectiveSchoolIdProvider** to lock each role to their authorized geographic scope.

Permission	State Official	District Officer	Block Officer	Field Inspector	School HM
View Schools	All	District	Mandal	Mandal	Own school
View Map	✓	✓	✓	✓	✓
AI + Manual Validation	✓	✓	✗	✗	✗
Field Inspection	✗	✗	✓	✓	✗
Export Data (PDF/Excel)	✓	✓	✗	✗	✗
Raise Demand	✗	✗	✗	✗	✓
District Filter	Free	Locked	Locked	Locked	Locked
Mandal Filter	Free	Free	Locked	Locked	Locked
Dashboard Tabs	6 tabs	6 tabs	5 tabs	5 tabs	2 tabs

Data Scoping Architecture

Role-based data scoping is implemented through Riverpod providers that lock geographic filters based on the current user's role:

- **effectiveDistrictProvider**: Locks District Officers, Block Officers, Field Inspectors, and HMs to their assigned district. State Officials can freely select any district.
- **effectiveMandalProvider**: Locks Block Officers, Field Inspectors, and HMs to their assigned mandal. State and District Officials can freely select mandals.
- **effectiveSchoolIdProvider**: Locks School HMs to their single assigned school. All other roles see multiple schools.

10. Offline-First Architecture

Vidya Nirmaan is designed for rural deployment where network connectivity is unreliable. The Hive-based offline cache provides 5 dedicated storage boxes for school data, demand plans, pending assessments, offline demand submissions, and sync metadata. Data flows through a try-online-first, fallback-to-cache strategy with automatic queue-and-retry for write operations.

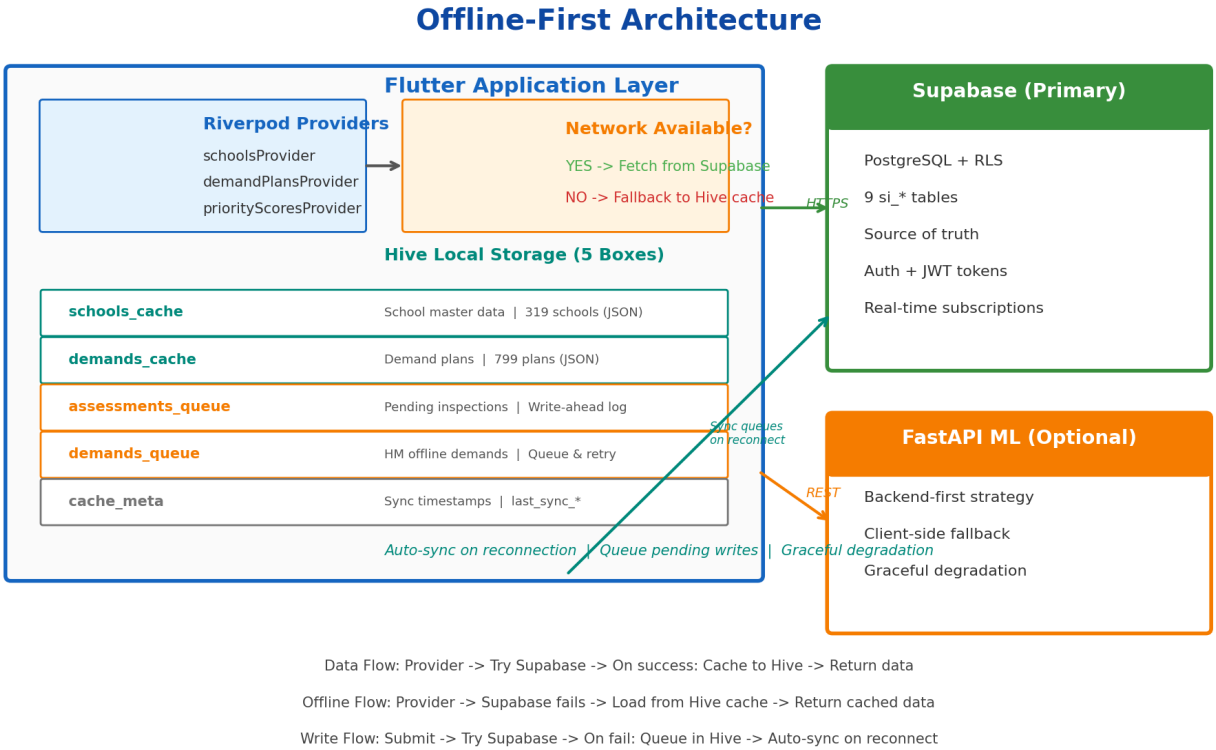

Figure 7: Offline-First Architecture — Hive Cache with Auto-Sync

11. Screen Navigation & User Flow

The application follows a role-adaptive navigation pattern. After role selection, the dashboard dynamically renders different tab sets based on the user's role: State/District Officials see 6 tabs including validation, Block Officers and Field Inspectors see 5 tabs with inspection capabilities, and School Head Masters see a purpose-built 2-tab interface focused on their single school and demand management.

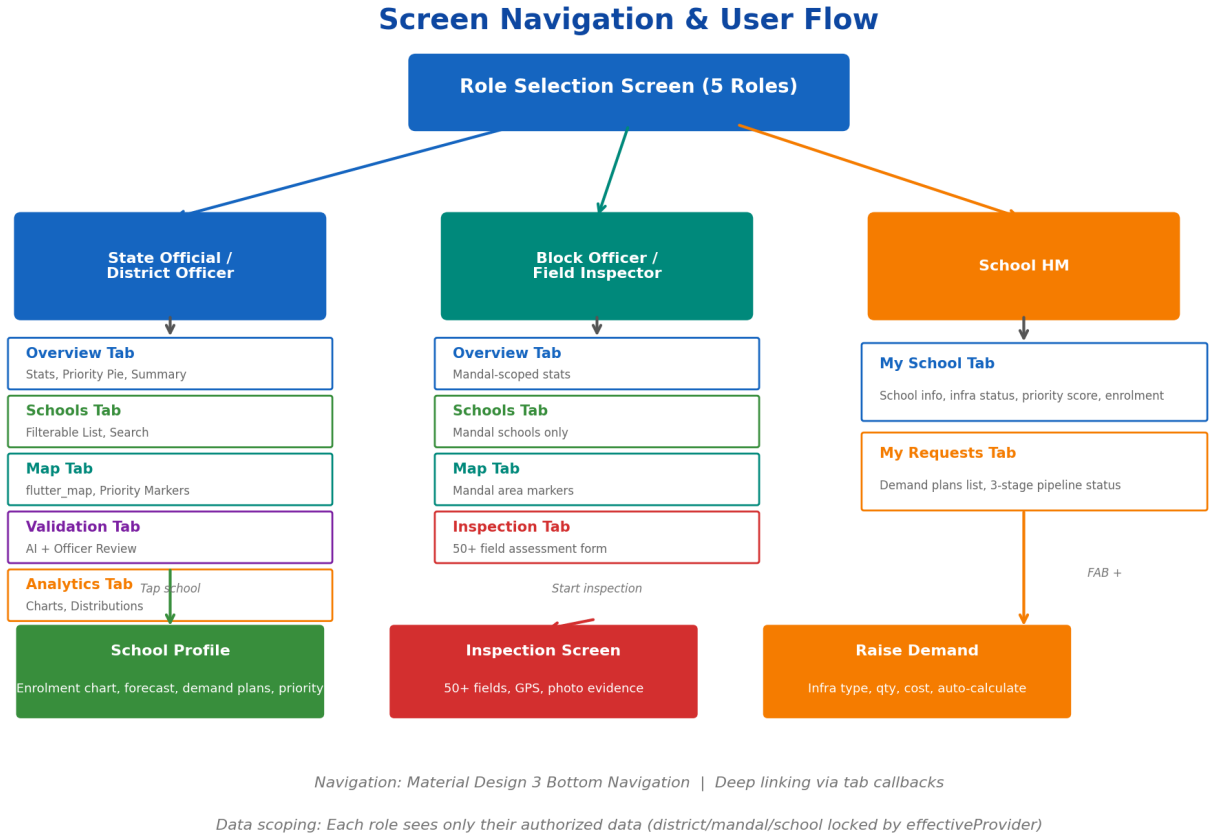

Figure 8: Screen Navigation — Role-Adaptive Dashboard with Detail Screens